CSE212: SOFTWARE DEVELOPMENT METHODOLOGIES

YEDITEPE UNIVERSITY

SPRING 2026

ASSIGNMENT 2 – DUE DATE MARCH 27TH, 2026

Below, you will find the description of your second assignment. Additionally, you can find the expected output of this assignment in the *Application Walkthrough* section.

As part of your second assignment, you will need to improve your existing *Online Shop Inventory Management System*. Your updated system must now allow a customer to place **an order** using the items currently stored in the inventory and calculate the total cost of that order.

Accordingly, you are required to create a new ***Order*** class with the following attributes:

- *orderID (int):* A value that must increase each time a new order is placed.
- *inventory (Inventory):* A reference to the ***Inventory*** object from which the customer places the order.
- *itemName (String):* The name of the item being ordered.
- *quantity (int):* The quantity of the selected item being ordered.

In the ***Order*** class, you are also expected to implement the following methods:

- *displayOrderInfo():* This method should display the details of the current order. The displayed information must include:
  - order ID,
  - item name,
  - quantity.
- *checkStorage():* This method should check whether the requested item exists in the inventory. The order should be allowed only if the item exists in the inventory. If the item does not exist, the program must display an appropriate message
- *calculateOrderCost():* This method should calculate the total cost of the order. The cost must be calculated by multiplying the price of the ordered item from the ***Inventory***, by the ordered quantity. The method should then display the calculated order cost.

You must improve your menu from Assignment 1 by adding customer order operations. Also, including this assignment onwards, you must implement your menu options using **enums**. Your updated program should support the following menu options:

INVENTORY MENU:

1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory

CUSTOMER MENU:

5. Add an order
6. Calculate order cost
7. Exit

For the purposes of this assignment, we only expect your program to support one active order at a time.

PS: As this is a follow up to your existing assignment (#1) your inventory data **must** be still stored in C-style arrays. Test your program thoroughly to ensure it meets the requirements of the assignment.

Application Walkthrough

```
INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
```

```
Enter your choice: 1
Enter item name: Iphone 15 Pro
Enter item price: $999
Enter item quantity: 3
Item added successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 1
Enter item name: AirPods Pro
Enter item price: $329
Enter item quantity: 2
Item added successfully.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 2
Inventory Items:
Item 1: Iphone 15 Pro, Price: $999.0, Quantity: 3
Item 2: AirPods Pro, Price: $329.0, Quantity: 2

INVENTORY MENU:
```

```
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 3
Total number of items: 2

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit

Enter your choice: 5
Enter the item name you want to order: AirPods Pro
Enter the quantity: 3
Item not found in inventory. Order cannot be placed.

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit
Enter your choice: 5
Enter the item name you want to order: Iphone 15 Pro
```

```
Enter the quantity: 1
Order ID: 2
Item Name: Iphone 15 Pro
Quantity: 1

INVENTORY MENU:
1. Add a New Item
2. Display all Items
3. Display the Total Number of Items
4. Calculate the Total Worth of the Inventory
CUSTOMER MENU:
5. Add an Order
6. Calculate Order Cost
7. Exit

Enter your choice: 6
Order cost for 1 Iphone 15 Pro is: $999.0
Exiting... Goodbye!

Process finished with exit code 0
```

Submit your assignment folder in a zip file [class files will not be marked**], which has your student number as name, through the YULEARN (https://yulearn.yeditepe.edu.tr) latest by the end of Friday, March 27th, 2026.**